Математические выкладки для задачи о "прощёлкивании" упругой арки.

Начальный профиль арки есть

$$y_0(x) = a_1^0 \sin\left(\frac{\pi x}{\ell}\right)$$

ПОД нагрузкой профиль арки описывается рядом Фурье:

$$y(x) = \sum_{j=1}^{\infty} a_j \sin\left(\frac{\pi j x}{\ell}\right)$$

В приближении двух гармоник

$$y(x) \simeq a_1 \sin\left(\frac{\pi x}{\ell}\right) + a_2 \sin\left(\frac{2\pi x}{\ell}\right)$$

Вычислим энергию изгиба арки:

$$E_б = \frac{EI}{2}\int_0^{\ell} (y''(x))^2 dx$$

$$y''(x) = -\left(\frac{\pi}{\ell}\right)^2 a_1 \sin\left(\frac{\pi x}{\ell}\right) - \left(\frac{2\pi}{\ell}\right)^2 a_2 \sin\left(\frac{2\pi x}{\ell}\right)$$

① $$\int_0^{\ell} \sin^2\left(\frac{\pi x}{\ell}\right) dx = \frac{1}{2}\int_0^{\ell} \left(1 - \cos\left(\frac{2\pi x}{\ell}\right)\right) dx =$$

$$= \frac{\ell}{2} - \frac{1}{2}\int_0^{\ell} \sin\left(\frac{2\pi x}{\ell}\right) dx = \frac{\ell}{2}.$$

$$\| \quad 0$$

② $$\int_0^{\ell} \sin^2\left(\frac{2\pi x}{\ell}\right) dx = \frac{1}{2}\int_0^{\ell} \left(1 - \cos\left(\frac{4\pi x}{\ell}\right)\right) dx =$$

$$= \frac{\ell}{2} - \frac{1}{2}\int_0^{\ell} \cos\left(\frac{4\pi x}{\ell}\right) dx = \frac{\ell}{2}.$$

$$0 \quad \| \quad 0$$

③ $\int_0^\ell \sin\left(\frac{\pi x}{\ell}\right)\sin\left(\frac{2\pi x}{\ell}\right)dx = \frac{1}{2}\left[\int_0^\ell \cos\left(\frac{\pi x}{\ell}\right)dx - \int_0^\ell \cos\left(\frac{3\pi x}{\ell}\right)dx\right] = 0$

$\int_0^\ell \cos\left(\frac{3\pi x}{\ell}\right)dx = 0$

Таким образом,

$$E_б = \frac{EI\ell}{2\cdot 2}\left[\left(\frac{\pi}{\ell}\right)^4 a_1^2 + \left(\frac{2\pi}{\ell}\right)^4 a_2^2\right] = \frac{EI\ell}{4}\left[\left(\frac{\pi}{\ell}\right)^4 a_1^2 + \left(\frac{2\pi}{\ell}\right)^4 a_2^2\right]$$

Работа силы F, приложенной к середине арки в точке $x = \ell/2$, есть

$$W_F = \int F\,dx = F\left(y(x=\ell/2) - y_0(x=\ell/2)\right) = F(a_1 - a_1^0)$$

Потенциальная энергия $U(a_1, a_2, F)$ в приближении двух гармоник:

$$U(a_1, a_2, F) = E_б + W_F = \frac{EI\ell}{4}\left[\left(\frac{\pi}{\ell}\right)^4 a_1^2 + \left(\frac{2\pi}{\ell}\right)^4 a_2^2\right] + F(a_1 - a_1^0)$$

Уравнение связи $a_1(a_2)$:

$$L = \int_0^\ell \sqrt{1+(y')^2}\,dx \approx \int_0^\ell \left(1 + \frac{1}{2}(y')^2 - \frac{1}{8}(y')^4\right)dx =$$

$$= \ell + \frac{1}{2}\int_0^\ell (y')^2 dx - \frac{1}{8}\int_0^\ell (y')^4 dx,$$

$$y' = \frac{\pi}{\ell} a_1 \cos\left(\frac{\pi x}{\ell}\right) + \frac{2\pi}{\ell} a_2 \cos\left(\frac{2\pi x}{\ell}\right),$$

① $\int_0^\ell \cos^2\left(\frac{\pi x}{\ell}\right) dx = \frac{1}{2}\int_0^\ell \left(1+\cos\left(\frac{2\pi x}{\ell}\right)\right) dx = \frac{\ell}{2}$

② $\int_0^\ell \cos^2\left(\frac{2\pi x}{\ell}\right) dx = \frac{1}{2}\int_0^\ell \left(1+\cos\left(\frac{4\pi x}{\ell}\right)\right) dx = \frac{\ell}{2}$

③ $\int_0^\ell \cos\left(\frac{\pi x}{\ell}\right)\cos\left(\frac{2\pi x}{\ell}\right) dx = 0$

Пояснение,

$\frac{\ell}{2}\cdot\frac{1}{2}\int_0^\ell (y')^2 dx = \left[\frac{1}{2}\left(\frac{\pi}{\ell}\right)^2 a_1^2 + \frac{1}{2}\left(\frac{2\pi}{\ell}\right)^2 a_2^2\right]\frac{\ell}{2} =$

$= \frac{\ell}{4}\left[\left(\frac{\pi}{\ell}\right)^2 a_1^2 + \left(\frac{2\pi}{\ell}\right)^2 a_2^2\right].$

Для вычисления $\int_0^\ell (y')^4 dx$:

$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

① $\int_0^\ell \cos^4\left(\frac{\pi x}{\ell}\right) dx = \int_0^\ell \underbrace{\frac{1}{2}\left(1+\cos\left(\frac{2\pi x}{\ell}\right)\right)}_{\cos^2\left(\frac{\pi x}{\ell}\right)} \underbrace{\frac{1}{2}\left(1+\cos\left(\frac{2\pi x}{\ell}\right)\right)}_{\cos^2\left(\frac{\pi x}{\ell}\right)} dx =$

$= \frac{1}{4}\int_0^\ell \left(1+2\cos\left(\frac{2\pi x}{\ell}\right) + \underbrace{\cos^2\left(\frac{2\pi x}{\ell}\right)}_{\frac{1}{2}\left(1+\cos\left(\frac{4\pi x}{\ell}\right)\right)}\right) dx =$

$= \frac{1}{4}\int_0^\ell \left[\frac{3}{2} + 2\cos\left(\frac{2\pi x}{\ell}\right) + \frac{1}{2}\cos\left(\frac{4\pi x}{\ell}\right)\right] dx =$

$= \frac{3}{8}\ell + \frac{1}{2}\int_0^\ell \cos\left(\frac{2\pi x}{\ell}\right) dx + \frac{1}{8}\int_0^\ell \cos\left(\frac{4\pi x}{\ell}\right) dx = \frac{3}{8}\ell.$

$\parallel 0 \qquad\qquad \parallel 0$

② Для нечётных степеней

$\int_0^\ell \cos^3\left(\frac{\pi x}{\ell}\right)\cos\left(\frac{2\pi x}{\ell}\right) dx = \frac{1}{4}\int_0^\ell \cos\left(\frac{2\pi x}{\ell}\right)\left[3\cos\left(\frac{\pi x}{\ell}\right) + \cos\left(\frac{3\pi x}{\ell}\right)\right] dx =$

$$\cos 3x = \cos(x+2x) = \cos x\cos 2x - \sin x \sin 2x =$$

$$= \cos x \underbrace{(\cos^2 x - \sin^2 x)}_{\cos 2x = \cos(x+x)} - \sin x \cdot 2\sin x\cos x =$$

$$= \cos x(\cos^2 x - (1-\cos^2 x)) - 2\cos x(1-\cos^2 x) =$$

$$= 2\cos^3 x - \cos x - 2\cos x + 2\cos^3 x = 4\cos^3 x - 3\cos x;$$

$$\Rightarrow \boxed{\cos^3 x = \frac{1}{4}\cos 3x + \frac{3}{4}\cos x}$$

$$= \frac{3}{4}\int_0^l \cos\left(\frac{\pi x}{l}\right)\cos\left(\frac{2\pi x}{l}\right)dx + \frac{1}{4}\int_0^l \cos\left(\frac{2\pi x}{l}\right)\cos\left(\frac{3\pi x}{l}\right)dx = 0$$

$$\textcircled{3}\quad \int_0^l \cos^2\left(\frac{\pi x}{l}\right)\cos^2\left(\frac{2\pi x}{l}\right)dx = \int_0^l \frac{1}{2}\left(1+\cos\left(\frac{2\pi x}{l}\right)\right)\times$$

$$\times \frac{1}{2}\left(1+\cos\left(\frac{4\pi x}{l}\right)\right)dx = \frac{1}{4}\int_0^l \left[1+\cos\left(\frac{2\pi x}{l}\right)+\cos\left(\frac{4\pi x}{l}\right)+\right.$$

$$\left.+\cos\left(\frac{2\pi x}{l}\right)\cos\left(\frac{4\pi x}{l}\right)\right]dx = \frac{l}{4}$$

$$\textcircled{4}\quad \int_0^l \cos\left(\frac{\pi x}{l}\right)\cos^3\left(\frac{2\pi x}{l}\right)dx = 0$$

Поэтому, связь между a_1 и a_2:

$$L = l + \frac{l}{4}\left[\left(\frac{\pi a_1}{l}\right)^2 + \left(\frac{2\pi a_2}{l}\right)^2\right] -$$

$$- \frac{l}{8}\left[\frac{3}{8}\left(\frac{\pi a_1}{l}\right)^4 + \frac{3}{2}\left(\frac{\pi a_1}{l}\right)^2\left(\frac{2\pi a_2}{l}\right)^2 + \frac{3}{8}\left(\frac{2\pi a_2}{l}\right)^4\right];$$

Отсюда следует связь $a_1 = a_1(a_2)$.

$L = l + \frac{l}{4}\left(\frac{\pi q^0}{l}\right)^2 - \frac{l}{8}\frac{3}{8}\left(\frac{\pi q^0}{l}\right)^4$, $\to$ Из нач. условия

Разложим $q_1 = q_1 - q_1^0 + q_1^0$, где $|q_1 - q_1^0| \ll q_1^0$

$$L \simeq l + \frac{l}{4}\left(\frac{\pi}{l}\right)^2\left[(q_1^0 + q_1 - q_1^0)^2 + 4q_2^2\right] -$$

$$- \frac{l}{8}\left[\frac{3}{8}\left(\frac{\pi}{l}\right)^4 (q_1^0 + q_1 - q_1^0)^4 + \frac{3}{2}\left(\frac{\pi}{l}\right)^2\left(\frac{2\pi}{l}\right)^2 (q_1^0 + q_1 - q_1^0)^2 q_2^2 + \right.$$

$$\left. + \frac{3}{8}\left(\frac{2\pi}{l}\right)^4 q_2^2\right] \simeq l + \frac{l}{4}\left(\frac{\pi q_1^0}{l}\right)^2 - \frac{l}{8}\frac{3}{8}\left(\frac{\pi q_1^0}{l}\right)^4 +$$

сокращаются

$$+ \frac{l}{4}\left(\frac{\pi}{l}\right)^2\left[2q_1^0(q_1 - q_1^0) + 4q_2^2\right] - \frac{l}{8}\left[\frac{3}{8}\left(\frac{\pi}{l}\right)^4 4(q_1^0)^3(q_1 - q_1^0) + \right.$$

$$\left. + \frac{3}{2}\left(\frac{\pi}{l}\right)^2\left(\frac{2\pi}{l}\right)^2 (q_1^0)^2 q_2^2\right] + \text{слагаемое более высок. порядка малости } O(q_2^4)$$

В первом приближении отсюда следует

$$(q_1 - q_1^0)\left[\frac{\pi^2}{2l} q_1^0 - \frac{3}{16}\frac{\pi^4}{l^3}(q_1^0)^3\right] \simeq q_2^2\left\{-\frac{\pi^2}{l} + \frac{l}{8}\frac{3}{2}\frac{\pi^4}{l^4}(q_1^0)^2\right\}$$

т.к. $\frac{q_1^0}{l} \ll 1$, $\Rightarrow$ Второе слаг. мало

$$(q_1 - q_1^0)\frac{\pi^2}{2l} q_1^0 \simeq -\frac{\pi^2}{l} q_2^2, \quad \Rightarrow$$

В первом приближении

$$\boxed{q_1^0 - q_1 \simeq \frac{2q_2^2}{q_1^0}} \quad (1)$$

В общем говоря, $q_1^0 - q_1 = \frac{2q_2^2}{q_1^0} + Cq_2^4$,

для нахождения коэффиц. при q_2^2 достаточно знать (1)

$$U = \frac{EJ}{4}\left(\frac{\pi}{\ell}\right)^4\left[q_1^2 + 4q_2^2\right] - F(q_1^\circ - q_1) \approx$$

$$\approx \frac{EJ}{4}\left(\frac{\pi}{\ell}\right)^4\left[(q_1^\circ + q_1 - q_1^\circ)^2 + 16q_2^2\right] - F(q_1^\circ - q_1) \approx$$

$$\approx \frac{EJ}{4}\left(\frac{\pi}{\ell}\right)^4 (q_1^\circ)^2 + \frac{EJ}{4}\left(\frac{\pi}{\ell}\right)^4\left[2q_1^\circ(q_1 - q_1^\circ) + 16q_2^2\right] -$$

$$-F(q_1^\circ - q_1) \approx \frac{EJ}{4}\left(\frac{\pi}{\ell}\right)^4 (q_1^\circ)^2 + \frac{EJ}{4}\left(\frac{\pi}{\ell}\right)^4\left[\underbrace{16q_2^2 - 2q_1^\circ \cdot \frac{2q_2^2}{q_1^\circ}}_{12q_2^2}\right] -$$

$$-F\frac{2q_2^2}{q_1^\circ} = \frac{EJ}{4}\left(\frac{\pi}{\ell}\right)^4 (q_1^\circ)^2 + \frac{EJ}{4}\left(\frac{\pi}{\ell}\right)^4 12q_2^2 - F\frac{2q_2^2}{q_1^\circ} =$$

$$= \frac{EJ}{4}\left(\frac{\pi}{\ell}\right)^4 (q_1^\circ)^2 + q_2^2\left[3\left(\frac{\pi}{\ell}\right)^4 EJ - \frac{2F}{q_1^\circ}\right];$$